

Excitement Projector: Augmenting Excitement-Perception and Arousal through Bio-signal-based Haptic Feedback in Remote-Sport Watching

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ABSTRACT

For non-sport fans, perceiving the excitement of surrounding fan groups and the arousal of collective emotions are some of the crucial factors that motivate their engaged excitement and loyalty in a sport; these factors are closely related to the process of evolution from a non-fan to a fan. The global COVID-19 pandemic changed the way of sport-watching from watching at the arena to watching from home alone. This has highlighted significant difficulties in the excitement transmission and arousal channel between non-sport fans and fans. Previous remote emotional intervening mediums had been limited to the use of virtual avatars to convey partners' external cues (such as appearance), to enhance the sense of presence from visual-audio perspectives. In this study, we explored a novel remote emotional intervening medium that conveys sport-fans' internal cues (bio-signals) that are widely believed to be related to internal emotional states of human, and displays those signals in a way that gives non-sport fans a deeper and more immersive experience: haptic feedback experience. Three bio-signal-based haptic feedback prototypes were developed, including heart-rate-based vibration, electromyography (EMG)-based pressure, and skin-temperature-based thermal feedbacks. An exploratory pilot study was conducted on a group of non-sport fans in a lab-control environment of remote-sport-watching to explore the effectiveness of the proposed mediums in enhancing their perception of sport-fan's excitement (emotion perception). Besides, we also analyzed non-sport fans' heart rate data when they were participants in the experiments to measure the performance of the proposed mediums in evoking the engaged excitement of non-sport fans (emotion arousal). Our results indicate the outstanding ability of EMG-based pressure feedback in effectively enhancing the process of emotion perception and the notable advantage of heart-rate-based vibration feedback in the arousal of non-sport fans' engaged excitement. This study presents the potential utility of bio-signal-based haptic feedback in augmenting remote emotional perception and arousal and also provides the underlining support for the future exploration and

development directions for social computing based on bio-signals and haptic technologies.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing**; **Haptic devices**.

KEYWORDS

Remote emotional perception, Remote emotional arousal, Bio-signals, Haptic feedback, Sport fandom, Remote-sport-watching

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1 INTRODUCTION

Intense competitive scenes in sports, such as the overtaking scenes in Formula One and scoring scenes in football matches, often invites the spectators' to experience a varied mixture of intense emotions. When shared among communities, this arouses collective emotions/effervescence [6], thereby establishing strong fan groups. Hatfield, Cacioppo, and Rapson [10] proposed the emotional contagion theory, which reveals the changes in the emotional trajectory experienced by the sport spectator from being a non-fan to a fan. Emotional contagion theory suggests that a self-described non-fan spectator is more likely to "capture" the excitement of the fan community around them (emotional perception), than to be motivated by engaged excitement and loyalty to the sport (emotion arousal) [8]. However, with the outbreak of Covid-19 globally, watching-at-home alone has been the most common method of sport-watching, which poses unique challenges in exploring an effective remote emotional intervening medium that can enhance non-sports fans' perception of collective excitement, which can further increase their engage excitement for the sport.

Most previous studies have developed emotional intervening mediums from visual and audio perspectives, which leveraged the virtual avatars in VR and AR environments to reproduce the external appearance (facial expression, gesture, head direction, etc.) or voice information of remote users to augment the social presence between remote users [11, 17]. However, given the limitation of human external cues in revealing their "inner feelings," and the inadequacy of visual and auditory feedback in providing an immersive

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emotional experience, we focused on exploring the possibility of effectively leveraging the internal cues of remote users to promote emotional perception and arousal.

Recently, the emergence of modern sensing technologies has provided opportunities to take advantage of personal bio-signals as cues to facilitate social connections [2]. Several recent works have validated that sharing personal bio-signals in the form of visual or auditory approaches, such as heart rate (HR), skin-conductance and temperature could help with improving emotional perception of, and connection with, others [5, 9, 13, 19]. Liu et al. [13] demonstrated that sending or feeding back animated avatars reflecting the heartbeats of couples on the Apple Watch could effectively enhance couples' emotional understanding and connection, thereby promoting positive interpersonal relationships. Winters, Walker, and Leslie [19] explored the effectiveness of an auditory heartbeat in eliciting empathy perception and responses among users. In addition, skin conductance and temperature are some of the primary indicators used to measure anxiety and tension in the autonomic nervous system. There have also been studies that used these two bio-signals to develop interaction interfaces and discussed their effectiveness in serving social connections. For example, Cheung et al. [4] designed an auditory interface to display skin conductance and temperature signals, and validated its effectiveness in helping detect anxiety in children. Curran et al. [5] investigated the performance of visualizing others' skin conductance and temperature to improve social understanding and empathy perception.

However, all the works cited above focused on displaying external or internal cues of remote users from a visual or auditory perspective. Haptic sensation, as one of the five senses with which humans perceive the world, is a way to communicate with users through physical touch, which can augment and provide deep, immersive human emotions through physical contact, such as feeling vibration, pressure, and thermal [15]. Providing pressure forces or thermal forearm has been validated to be effective in augmenting positive emotional perception [7, 14]. Werner, Wettach, and Hornecker [18] validated that sharing a partner's heartbeat through vibration feedback can evoke the feeling of being very close to the partner.

Inspired by these previous works, we set out a new challenge in this study. We proposed a novel remote emotion-intervening medium, based on displaying bio-signals of remote users in their expression of haptic sensation. That is, bio-signal-based haptic feedback. We attempted to explore the following questions: (1) Can the sharing of sport-fan's bio-signals in the form of haptic feedback to non-sport fans effectively enhance non-fans' perception of fans' excitement (remote emotional perception)? (2) Whether and how the proposed medium can evoke non-fans' excitement in sports (remote emotional arousal)?

To answer these questions, we implemented several tasks in this work: (1) Three types of bio-signal-based haptic feedback prototypes were designed and developed: vibration feedback that simulates a dynamic change in the heartbeat of the fan when watching sport; pneumatic pressure feedback that simulates the electromyography (EMG) activities (such as clenching fist due to being in a state of tension) in the muscles of the fan's forearm; and thermal feedback based on the change in skin temperature in the palm of the fan. (2) An exploratory experiment was conducted on a group

of Formula One (F1) non-sport fans, because F1 is one of the most watched sports events in the world, yet it is quite specialized in its own way and has a specific audience. The high-speed, fierce overtaking that occurs within a few seconds often evokes strong emotions in fans, but it is confusing for non-fans to understand what the sport is entails. Therefore, non-fans of F1 are suitable subjects for investigating the effectiveness of the proposed methods. A self-report-based evaluation was carried out to validate the ability of each proposed haptic feedback generated, based on F1 fans' bio-signals, to enhance non-fans' perception of F1 fans' excitement, compared with watching sports videos without feedback. (3) The HR data of non-fans when they participated in the experiment were also measured to determine the influence of each proposed intervening medium in arousing the engaged excitement of non-fans in sports.

2 BIOSIGNAL-BASED HAPTIC FEEDBACK PROTOTYPING

2.1 Heart-Rate-based Vibration Feedback

The heartbeat was monitored using a pulse sensor (SFE-SEN-11574) that measured blood flow at an average frequency of 125 Hz, as shown in Figure 1 (a). Then, two HR-related features were derived to characterize the speed and strength of the heartbeat: the inter-beat interval (IBI), which measures the interval time between two beats, and the amplitude (Amp) of each beat. We employed a sine wave with the frequency of 100Hz as a baseline wave and applied real-time IBI and Amp values to the baseline wave to generate a continuous vibration waveform that enables a visualization of the dynamic changes in HR, as shown in Figure 1 (b). Voice coil motor (VCM) actuators were adopted to create heartbeat-based vibration feedback, which was driven by an amplifier with an average output power of 40 W, as shown in Figure 1 (c). To design a non-awareness feedback device that can be used in daily living without too much obtrusiveness, a neck pillow was employed as a feedback container. We fixed the VCM actuators into the parts of the pillow that closed the area between the wearer's chest and shoulder, this created a feeling of "as if the other person's heartbeat is very close to me".

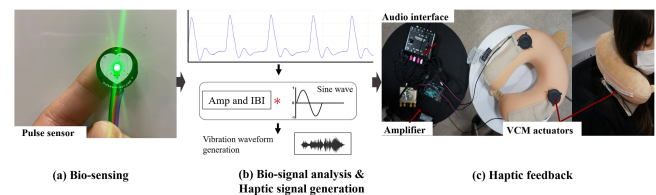


Figure 1: The prototyping of heart-rate-based vibration feedback

2.2 EMG-based Pressure Feedback

We leveraged the state that fans are nervous about the current race content and clench their fists as another mediating cue. The action of a clenching fist can be captured using electromyography measurements. We adopted the MyoWare Muscle Sensor to monitor the electrical activities of the muscle in the fan's forearm, at an average

frequency of 1000 Hz, as shown in Figure 2 (a). Figure 2 (b) presents the process of EMG data analysis. We first extracted the root mean square (RMS) of the raw EMG signals with a sliding window to smoothen the data. Considering that a larger window size would delay the detection response, while a smaller window size would lower the detection results owing to the distorted information, a sliding window length of 500 ms with a sliding step length of 250 ms was selected. We then integrated the average amplitude of the smoothed EMG value for each segmented window and used it as an EMG-related feature to indicate the contraction of muscles on the fan's forearm.

A self-developed haptic pressure force generator device, driven by pneumatic actuation, was employed as the pressure force feedback-providing device, as shown in Figure 2 (c). This device is composed of three substrates, including a power module, an FPGA drive module, and a drive module, which are used to generate compressed air with a pressure force ranged from 0-20 kPa by activating the built-in pump, valve, and pressure sensor. The value of the pressure force can be specified by sending a predesigned command via Bluetooth. A wristband-styled device made of nylon material was designed to contain the generated air; there are four small air chambers (1.5 cm × 3.5 cm for each) on the center of the wristband as shown in Figure 2 (c). Considering that higher pressure forces lead to more time consumption in retrieving the injected air, to simulate moment-by-moment muscle contraction activities as accurately as possible (while ensuring a threshold pressure force that can trigger nervousness in humans by generating 'gasping'), in our work, a pre-user-test survey was conducted in advance, and the pressure was finally set to 7 kPa.

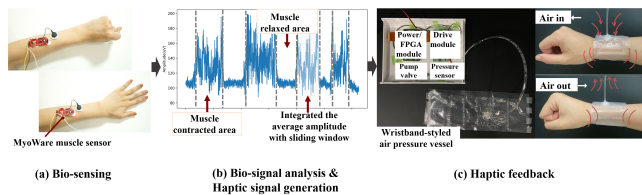


Figure 2: The prototyping of EMG-based pressure feedback

2.3 Skin-temperature-based Thermal Feedback

Increased skin temperature is an important cue that indicates the activation of automatic nervous system [12]. We thus utilized the skin temperature in the palm of the fans as another bio-signal cue. As shown in Figure 3 (a), a temperature sensor (LM35DZ) was attached to the palm for real-time skin temperature monitoring. Since providing thermal feedback on forearm could facilitate positive emotional generation [3, 14], in addition, increasing the temperature with 3-6 degree from the baseline 32°C could augment the excitement state of human [1]. We selected a Peltier element (TEC1-12706) driven by a MOS-FET module (2SK4017(Q)) as the thermal actuator. As shown in Figure 3 (b), the Peltier element was fixed to the upper area of forearm, where the warm stimuli side was closely attached to the inside of the forearm. Once the increased of fan's palm temperature was detected, a warm stimuli of 32 + 3°C was provided to the non-fan's inside forearm (see Figure 3 (c)). As the same

time, the surface temperature of the Peltier element was monitored in real-time through a temperature sensor; in case of excessive temperature increase due to a long heat release time, a cooling command will be delivered immediately once the temperature exceeds 32 + 6°C.

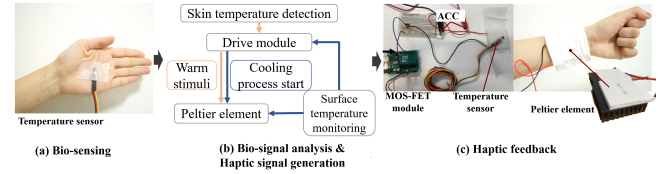


Figure 3: The prototyping of skin-temperature-based thermal feedback

3 PILOT STUDY

3.1 Participants and Procedure

A pilot study was conducted on seven self-identified non-fans (two females and five males, ranging in age from 30 to 45 years) who reported little or no experience watching F1 races. Considering that recruiting several fans or different race clips would introduce a bias in the results. We thus recruited one self-identified fan (female, age 26) who reported more than 10 years of experience watching F1 races with high loyalty to a specific driver, and adopted one video clip, which is recognized as the driver's most highlighted race moment and has received the most views on YouTube. This video clip is 2 min in length and contains the three best overtaking maneuvers honored in this driver's career so far.

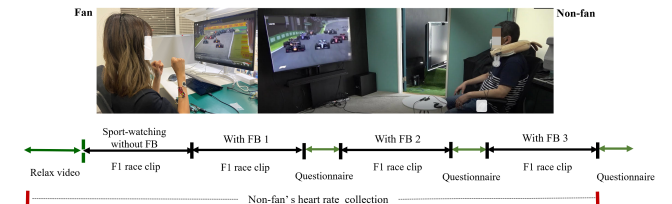


Figure 4: User test

Figure 4 presents the scenario of our user-evaluation experiment, where the F1 fan was watching the video clip while her bio-signals were monitored. The figure on the right side shows the non-fan watching the F1 clip while accepting the vibration haptic feedback that simulated the dynamic change in the heartbeat of the fan when she was watching the video previously. As shown at the bottom of Figure 4, the validation experiments included the following steps: (1) First, a rest time before the experiment was set where the non-fan was asked to watch a 1-min-long relaxing video. (2) The non-fan was asked to watch the race clip without receiving any feedback. (3) Next, the non-fan was requested to watch the video while receiving the proposed feedback. To avoid the influence of the feedback order on the results, feedback was provided randomly for each non-fan. Between each clip-watching period, non-fan needed to answer a

questionnaire based on a five-point scale (Strongly Disagree, Disagree, Neither Agree nor Disagree, Agree, and Strongly Agree) that contains two questions: (A) *Do you think the perception of fan's excitement has been enhanced by receiving the current feedback?* (B) *Do you think your excitement in this race has been augmented by receiving the current feedback?* A free description of the impression of non-fans on the current feedback was also requested. (4) The HR of the non-fans during the experiments was collected for further quantitative analysis. The experiment was approved by the Institutional Review Board (IRB) at that company.

3.2 Qualitative Analysis

The Results of the questionnaire, as shown in each row of Figure 5, confirm that 86% (see Figure 5 (b)) of non-fans agreed that accepting an air pressure feedback that simulates the fan's forearm muscle contraction effectively improved their perception of the fan's excitement. Simultaneously, a very similar percentage (72%, see Figure 5 (a)) of non-fans agreed that sensing the vibration feedback generated based the fan's HR helped enhance their perception of the fan's excitement. However, as shown in Figure 5 (c) only 14% of non-fans' agreed that their perception of fans' excitement was increased by the thermal feedback provided; in contrast, 43% of the non-fans took a neutral stance, while the remaining 43% clearly indicated that the thermal feedback did not provide any assistance in sensing the excitement state of fans. These results suggest the effectiveness of the proposed heart-rate-based vibration feedback and EMG-based pressure feedback in enhancing non-fans' perception of the fan's excitement. In addition, it was confirmed that EMG-based pressure could better do this job.

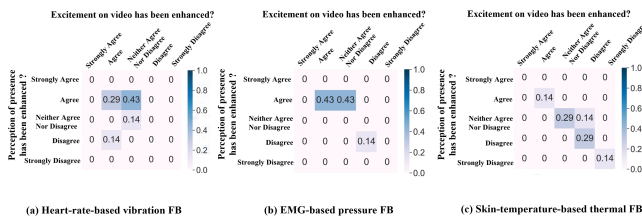


Figure 5: Results of qualitative analysis

Furthermore, we report the results of non-fans reporting a change in their feeling of engaged excitement in race. 43% (see Figure 5 (a)) of the non-fans approved of the ability of heart-rate-based vibration feedback to evoke their engaged excitement while watching the F1 race, regardless of some of them denying its effectiveness in enhancing the perception of the excitement of the fan. Other non-fans expressed ambiguous opinions on the role of this feedback. In addition, the same percentage of non-fans (43%, see Figure 5 (b)) experienced an enhancement in engaged excitement while watching the F1 race when receiving air pressure feedback. However, different from the results regarding the vibration feedback, 14% of non-fans expressed a negative evaluation of the effectiveness of air pressure feedback. However, up to 53% (see Figure 5 (c)) of the non-fans reported ineffectiveness of thermal feedback in arousing any engaged excitement to watch this race.

In addition, most non-fans expressed the following regarding each haptic feedback:

"I could sense the fan was feeling tension and excitement to the current scene when the pressure feedback suddenly grasped my wrist, and the moment I was 'grasped,' I began to realize that this scene should be a highlight moment that I need to notice."

"I think I can feel the emotions of the fan change with the change of the sport-scene through the vibrating feedback, such as, in the overtaking scene, the vibration became stronger, faster, more changes, which makes me sense that the fan was very excited, and I was infected by this strong emotion and started to be excited."

“For the thermal feedback, I felt the change of the temperature on my forearm but I can’t associate this change with fan.”

"The vibration can be sensed during the whole sport-video, I started getting used to this vibration after a while, while, Pressure feedback is instantaneous, fast and short-lived, I think my emotions can be more driven by pressure feedback than vibration feedback."

These results provide more detailed evidence about the effectiveness of heat-rate-based haptic feedback and EMG activity based haptic feedback to enhance the non-fan's perception of the fan's excitement. However, further quantitative analysis is required to explore the effectiveness of different proposed feedbacks in augmenting the engaged excitement of non-fans for a sport.

3.3 Quantitative Analysis of Non-fans' Heart Rate

Quantitative validation was performed by analyzing the HR and heart rate variability (HRV) of non-fans who participated in the experiments. As shown in Figure 6, a series of HR- and HRV-related features from the time-domain: the standard deviation of NN intervals (SDNN), root mean square successive difference (RMSSD), mean HR and frequency domain (low frequency band (LF: 0.04-0.15 Hz); high frequency band (HF: 0.15-0.4 Hz)), and LF/HF ratio were derived. Among them, the index of SDNN, mean HR, LF, and LF/HF are closely related to the sympathetic nervous system (SNS) and are positively correlated with human excitement, whereas RMSSD and HF are related to the parasympathetic nervous system (PNS) and negatively correlated with human excitement [16].

As shown in the Figure 6 (a), both the heart-rate-based vibration feedback and the EMG-based pressure feedback showed excellent ability to arouse excitement of non-fans, with an increase in the average SDNN and average mean-HR and decrease in average RMSSD compared with values when no feedback was provided. Moreover, the heart-rate-based vibration feedback did a better job than EMG-based pressure feedback in helping with improved excitement level of non-fans, increasing the SDNN value by 25%, compared with the pressure feedback increasing the SDNN value by 20.9%. Similar results were observed from the frequency-domain features as shown in the Figure 6 (b). All the three proposed methods presented outstanding capability in enhancing the excitement level of non-fans. Among them, heart-rate-based vibration feedback achieved the best performance and yielded a mean LF/HF ratio of 1.26, stronger than the EMG-based pressure feedback with a mean LF/HF of 1.20 and better than the thermal feedback with a mean LF/HF of 1.14. These

exciting results indicate that, all the proposed haptic feedback signals could help enhance the engaged excitement of non-fans in a sport; furthermore, it was found that vibration feedback can be used to better evoke non-fan's excitement in a sport.

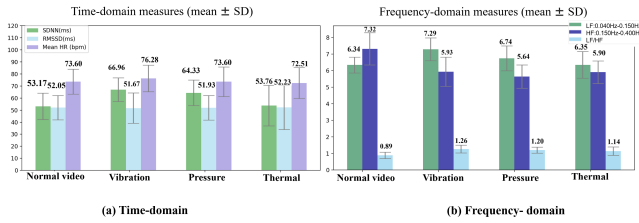


Figure 6: Results of quantitative analysis of non-fans' heart rate

4 CONCLUSION AND FUTURE WORK

4.1 Summary

In this study, we proposed a novel remote emotion-intervening medium that shares sport-fan's bio-signals (in the form of haptic expressions) with non-sports fans, in order to enhance non-fans' perception of fan's excitement during remote sport-watching, and to arouse non-fans' engaged excitement in a sport. Three types of bio-signal-based haptic feedback were designed and developed, including a heart-rate-based vibration feedback that can reproduce dynamic changes in the heartbeat of the fan when watching sports; an air pressure feedback generated based on the EMG activities in a fan's forearm that transmits the muscle contraction caused by tension when the fan is watching the sport, such as a clenching fist; and a thermal feedback that characterizes the change in skin temperature in the palm of a fan's hand. An exploratory experiment was conducted on a group of F1 non-fans to explore the performance of the proposed bio-signal-based haptic feedback in enhancing non-fans' perception of the fans' excitement in remote sport-watching and to understand how different proposed intervening mediums influence the arousal of non-fans' excitement in sports.

Qualitative analysis was performed based on non-fans' self-report, and the results suggest the effectiveness of the heart-rate-based vibration feedback and EMG-activity-based pressure feedback in enhancing non-fans' perception of fans' excitement in remote sport-watching. The pressure feedback could better convey the states of "tension" and "excitement" felt by fans. At the same time, while experiencing the changes in vibration, non-fans could feel the emotions of the fans changing with the changes in sports scenes, thereby arousing their enthusiasm for the sport.

HR data of the non-fans were also collected and analyzed, and a series of HR-related features from both the time-domain and frequency-domain were derived to indicate the change in excitement of the non-fans under the acceptance of each proposed bio-signal-based haptic feedback. The results suggest that heart-rate-based vibration feedback can perform better in evoking non-fans' excitement in a sport compared to the other two options in this study.

4.2 Limitations and Future Work

There are several limitations we would like to point out for future improvement and exploration: (1) The current user study was performed on a small group of F1 non-fans; we would like to conduct a further validation on a larger scale, and explore the performance of the proposed methods in different sports. (2) This work discusses the usage of the proposed emotion intervention medium in a uni-directional emotion perception and transmission task, which conveys the sport-fan's emotion to the non-sport fan. We are also interested in revealing the performance of the proposed methods in a bi-directional emotion connection, in which the aroused emotions of non-fans are sent back to fan groups and to explore the mechanism of this bi-directional bio-signal-based haptic feedback with social computing. (3) The current work proposed 3 independent systems and investigated their abilities respectively in excitement perception among fans and non-fans respectively, in our next work, we would like to implement a fusion system that can provide multiple bio-signal-based haptic feedbacks and further explore its performance in both uni- and bi-directional emotion connection.

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